

Project Title: CampusSwap

Project Members:

- Elif Doğa Tarhana
- Berkant Acun
- Uğur Pehlivan
- Melih Kaan Beşir
- Ahmet Emir Civelek

Scenario:

Overview: CampusSwap is a web-based system for university students to exchange academic resources. Users can register and log in to the system. After logging in, they can create, update, and delete posts to share resources. To find materials, users can browse and search posts, and they can filter these posts by department. Also, students can send exchange requests to other users. Users can manage these requests by accepting or rejecting them. The main goal of this system is to make sharing educational resources easy and organized for all students.

Entities and Primary Keys:

User: Members of the platform have a unique `user_id`, username, email, password, and an avatar name. Each user is also associated with a specific department through a `dept_id`. The **`user_id`** uniquely identifies each user.

Resource: This entity represents the materials being shared. Each resource has a unique `resource_id`, a type (e.g., PDF, book), and a description. It is also categorized by a `dept_id`. The **`resource_id`** is the key attribute.

Post: Users create posts to describe what they are offering and what they are requesting in return. A post includes a unique `post_id`, the `user_id` of the owner, an `offer_id`, a `request_id` and the current status of the listing. The **`post_id`** is the unique identifier.

Exchange: This entity manages the trade requests. It tracks has a unique `exchange_id`, the related `post_id`, the `requester_id` (the user asking for the trade), and the transaction status. The **`exchange_id`** uniquely identifies each exchange attempt.

Department: To organize users and resources, the system stores department information. Each department has a unique `dept_id`, a name, and belongs to a specific faculty. The **`dept_id`** is the key attribute.

Relationships:

A user can create multiple posts; but each post belongs to exactly one owner(1:N). A post links to specific resources; many posts can use the same resource from the system(N:1). Multiple users can request a trade for a single post; only one can be accepted(1:N). One department can have many users ; one user can have one department(1:N). A department can have many resources ; a resource can have one department(1:N). Multivalued Attributes: A user can have multiple skills at the same time.